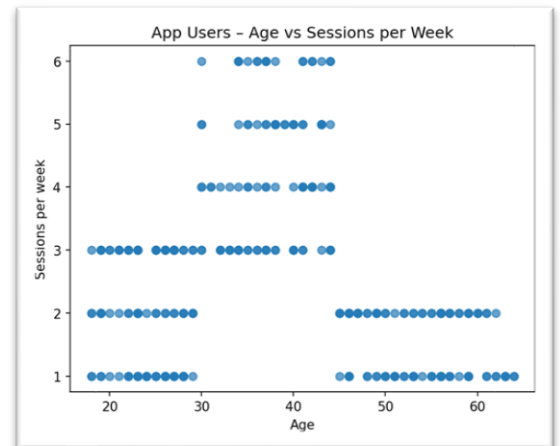


Behaviour-Based Segmentation of App Users Using K-Means Clustering

1. Background & Objective

Our app serves a broad user base with very different usage patterns. Some users log in frequently and stay for long sessions, while others only check in briefly once in a while. Today, most engagement and marketing activities are still “one-size-fits-all”, which risks over-communicating to some users and under-serving others.

Objective: Use data-driven segmentation to group users by behaviour, so we can design targeted engagement strategies for each segment instead of treating all users the same.



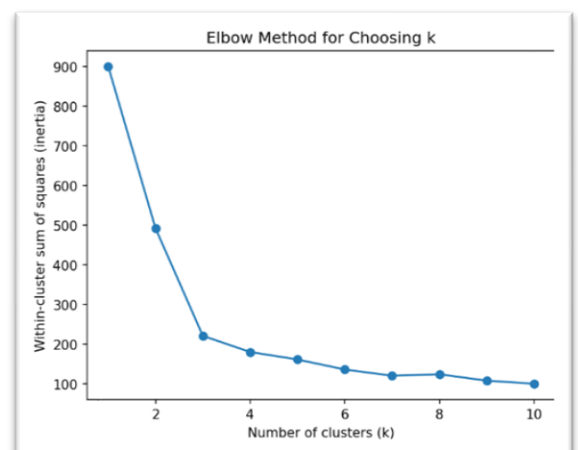
2. Data & Method

We used anonymised user-level data with three key behavioural features:

- **Age**
- **Sessions per week**
(how often users log in)
- **Average session duration**
(how long they stay per visit)

The steps were:

1. **Standardise features** so that age, frequency and duration are on a comparable scale.
2. Apply **K-Means clustering** to group users based on these three behaviours.
3. Use the **elbow method** to choose the number of clusters by testing $k = 1-10$ and observing where additional clusters give diminishing returns.

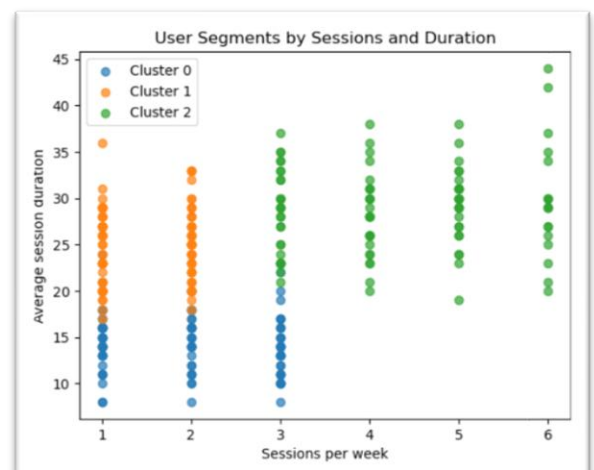


The elbow point appears around $k = 3$, which balances model simplicity with meaningful separation between groups.

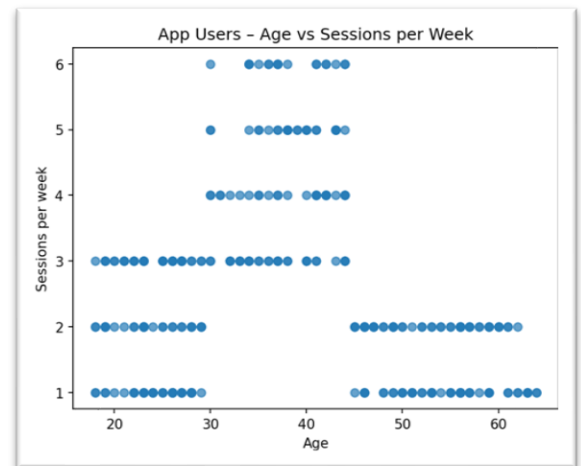
3. Key Findings: Three Behavioural Segments

K-Means with $k = 3$ reveals three distinct behaviour-based segments:

- **Cluster 0 – Light, quick users**
Fewer sessions per week and shorter session durations.
These users dip in and out quickly.



- **Cluster 1 – Occasional but deep users**
1–2 sessions per week but relatively long sessions when they log in. They explore more when they do come.
- **Cluster 2 – Heavy, highly engaged users**
Highest number of sessions per week and longest session durations. These are our power users.



4. Business Recommendations

The segments suggest different engagement playbooks:

- **Cluster 0 – Light, quick users (Nurture)**
 - Use gentle, low-effort nudges (simple reminders, quick tips) to increase visit frequency.
 - Help them build a basic usage habit without overwhelming them.
- **Cluster 1 – Occasional but deep users (Deepen)**
 - Highlight richer features and content that reward longer sessions (advanced tools, long-form content, personalised recommendations).
 - Encourage a modest increase in visit frequency to unlock more value.
- **Cluster 2 – Heavy, highly engaged users (Retain & Leverage)**
 - Prioritise retention with loyalty rewards and early access to new features.
 - Involve them in feedback loops and referral programmes; they are strong candidates for advocacy.

5. Impact & Next Steps

By moving from a generic approach to a **behaviour-based segmentation strategy**, we can:

- Target different user groups with tailored messages and offers.
- Run A/B tests by segment to measure uplift in retention, sessions per week and revenue per user.
- Gradually embed these segments into ongoing marketing and product decisions.

This provides a practical, data-backed foundation for increasing engagement and lifetime value across the user base.

